Steering in the Expanded MLP Space: Activation Intervention in the 4× Wider Intermediate Representation

Ari Holtzman and NeuriCo

Abstract

Activation steering—adding a direction vector to a model's internal representations at inference time—is a simple and effective technique for controlling language model behavior. Nearly all prior work applies steering vectors to the residual stream, a d_{model}-dimensional space that aggregates information from all model components. We ask whether the MLP intermediate space, a $4\times$ wider post-activation representation where the model performs much of its computation, is a better or worse target for steering. We conduct a systematic comparison across three intervention points in GPT-2 Small: the residual stream (768-dim), the MLP intermediate space (3072-dim, post-GELU), and the MLP output (768-dim, after down-projection). We find that steering in the MLP intermediate space is possible but approximately $2.6\times$ less efficient than residual stream steering, as measured by sentiment probability shift per unit of KL divergence. Linear probes achieve lower accuracy in the MLP space (76%) than the residual stream (86%), contradicting the hypothesis that the wider space is more linearly separable. However, the MLP steering vector is remarkably sparse: just 2.8% of neurons carry 50% of the steering signal. This concentration makes MLP steering fragile—generations degrade to gibberish at lower intervention strengths. Our results suggest that the residual stream's advantage stems from its role as an integrative representation, while MLP neurons encode specific, less universal features that resist broad behavioral steering.

1 Introduction

Activation steering lets us control what a language model says without changing its weights. The idea is simple: compute a direction in activation space that corresponds to a concept (say, positive sentiment), and add a scaled version of that direction during inference. This approach, popularized by Turner et al. [2023] and Zou et al. [2023], works well and requires no training. But where in the network should we intervene?

Nearly all steering work targets the *residual stream*—the d_{model}-dimensional representation that flows through the transformer, accumulating contributions from attention heads and MLP sublayers. This is a natural choice: the residual stream is the model's "main highway," and a single perturbation there affects all downstream computation.

But the residual stream is not the only option. Inside each MLP sublayer, activations pass through an intermediate space that is $4\times$ wider ($d_{\text{mlp}} = 4 \cdot d_{\text{model}}$). After the up-projection and nonlinearity, the model operates in a 3072-dimensional space (for GPT-2 Small) before projecting back down to 768 dimensions. This expanded space is where MLP neurons act as feature detectors [Cunningham et al., 2023], and recent work shows it can be decomposed into interpretable, sparse features [Dunefsky et al., 2024, Shafran et al., 2025]. A natural question arises: *is the MLP intermediate space a better target for activation steering?*

There are reasons to think it might be. The space is wider, so concepts may be less superimposed. Post-activation sparsity (from GELU/ReLU) could provide a more natural basis for intervention. And Shafran et al. [2025] show that MLP-space features derived via semi-nonnegative matrix factorization outperform residual-stream baselines at causal steering—though they use learned features rather than simple mean-difference vectors.

We test this hypothesis directly. We compute mean-difference steering vectors [Turner et al., 2023] in three spaces within GPT-2 Small—the residual stream, the MLP intermediate (post-GELU) space, and the MLP output—and compare their effectiveness at steering sentiment. We evaluate steering efficiency (sentiment shift per unit of KL divergence), linear separability (probe accuracy), and the sparsity structure of each space.

Our main findings are:

- We show that mean-difference steering in the MLP intermediate space is feasible but $2.6\times$ less efficient than residual stream steering, as measured by sentiment probability shift per unit of KL divergence.
- We find that the MLP intermediate space is *less* linearly separable than the residual stream (76% vs. 86% probe accuracy), contradicting the intuition that a wider space should be more linear.
- We characterize the sparsity structure of MLP steering vectors: just 2.8% of neurons (86 out of 3072) carry 50% of the steering signal, compared to 43% of dimensions needed in the residual stream.
- We demonstrate that this concentration makes MLP steering fragile—steered generations degrade to incoherence at lower intervention strengths than in the residual stream.

These results clarify when and why the MLP intermediate space is a useful intervention target. The residual stream is more efficient for general-purpose steering because it integrates information from all model components. The MLP space, by contrast, encodes specific features that concentrate the steering signal in few neurons—useful for targeted interventions, but fragile under the blunt instrument of mean-difference steering.

2 Related Work

Activation steering. Turner et al. [2023] introduce activation addition (ActAdd), computing steering vectors as mean activation differences between contrastive prompt pairs and adding them to the residual stream. Zou et al. [2023] extend this idea under the umbrella of representation engineering, using linear probes and interventions for top-down model control. Several works have refined these methods: Rimsky et al. [2024] scale steering to safety-relevant behaviors, and more recent work explores conditional activation steering [Lee et al., 2024], multi-property steering [Wang et al., 2024], and angular steering [Wang et al., 2025]. A key finding from Tan et al. [2024] is that steering vectors have substantial reliability limitations—their effectiveness varies across inputs and can reflect spurious correlations. All of these methods operate in the residual stream; we extend the comparison to the MLP intermediate space.

MLP interpretability. The MLP sublayer performs a large fraction of a transformer's computation, yet its intermediate space remains less studied than the residual stream. Cunningham et al. [2023] train sparse autoencoders (SAEs) on model activations and find that the learned features are more interpretable than individual neurons. Dunefsky et al. [2024] introduce transcoders—wide, sparse approximations of MLP sublayers that map input to output directly—enabling input-invariant circuit analysis through the MLP. Most relevant to our work, Shafran et al. [2025] decompose MLP intermediate activations using semi-nonnegative matrix factorization (SNMF) and show that the resulting features outperform both SAEs and supervised baselines (DiffMeans) at causal steering. Their features reveal hierarchical, compositional structure where related concepts share core neurons. Marks et al. [2026] provide further evidence that MLP computations are sparse in the neuron basis, with only a small fraction contributing to any given task.

Linear representations in neural networks. The linear representation hypothesis [Nanda et al., 2023, Park et al., 2024] posits that high-level concepts are encoded as linear directions in activation space. This hypothesis underpins both linear probing and activation steering. If concepts are indeed linear, then mean-difference vectors should align with concept directions, and wider spaces with less

superposition should be easier to steer. We test this prediction by comparing linear probe accuracy across spaces.

Our contribution. Prior work on MLP-space steering uses learned features (transcoders, SNMF) rather than direct mean-difference vectors. We ask the simpler question: does the basic mean-difference approach—the workhorse of activation steering—work better or worse in the MLP intermediate space? This comparison isolates the effect of the intervention space from the effect of the feature extraction method.

3 Methodology

3.1 Model and Intervention Points

We use GPT-2 Small [Radford et al., 2019] (124M parameters, 12 layers, $d_{\text{model}} = 768$, $d_{\text{mlp}} = 3072$) via TransformerLens [Nanda and Bloom, 2022]. We compare three intervention points at layer 8, chosen as a middle layer where representations are sufficiently processed:

1. **Residual stream** (RESID): `blocks.8.hook_resid_post`, 768-dim. The standard intervention target that integrates attention and MLP contributions.
2. **MLP intermediate** (MLP-INTER): `blocks.8.mlp.hook_post`, 3072-dim. The post-GELU expanded representation inside the MLP sublayer.
3. **MLP output** (MLP-OUT): `blocks.8.hook_mlp_out`, 768-dim. The MLP's contribution after down-projection by $\boldsymbol{W}_{\text{out}}$, before addition to the residual stream.

3.2 Steering Vector Construction

We follow the mean-difference method of Turner et al. [2023]. Given a set of positive-sentiment prompts $\{x_i^+\}_{i=1}^{25}$ and negative-sentiment prompts $\{x_i^-\}_{i=1}^{25}$, we collect activations $\boldsymbol{h}_i^+$ and $\boldsymbol{h}_i^-$ at each intervention point (at the last token position) and compute the steering vector:

$$\boldsymbol{v}_{\text{steer}} = \frac{\bar{\boldsymbol{h}}^+ - \bar{\boldsymbol{h}}^-}{\|\bar{\boldsymbol{h}}^+ - \bar{\boldsymbol{h}}^-\|_2}, \quad \text{where} \quad \bar{\boldsymbol{h}}^\pm = \frac{1}{25}\sum_{i=1}^{25} \boldsymbol{h}_i^\pm. \tag{1}$$

The vector is L2-normalized so that the scaling factor α controls the intervention magnitude on a common scale across spaces.

3.3 Steering Application

During inference, we add the scaled steering vector to activations at the corresponding hook point:

$$\boldsymbol{h}' = \boldsymbol{h} + \alpha \cdot \boldsymbol{v}_{\text{steer}}. \tag{2}$$

We test $\alpha \in \{0, 5, 10, 20, 50, 100, 200, 500\}$. Positive α steers toward positive sentiment.

3.4 Prompt Design

We construct three prompt sets:

- **Positive** (25 prompts): Express clear positive sentiment (e.g., "I absolutely love this beautiful day, everything feels wonderful and").
- **Negative** (25 prompts): Express clear negative sentiment (e.g., "I absolutely hate this terrible day, everything feels awful and").
- **Neutral** (20 prompts): Sentiment-ambiguous inputs for evaluation (e.g., "The weather today is").

3.5 Evaluation Metrics

Sentiment probability shift. We define sets of 30 positive words ($\mathcal{P}$: *good, great, excellent, wonderful*, ...) and 30 negative words ($\mathcal{N}$: *bad, terrible, horrible, awful*, ...). For each neutral prompt, we measure the change in mean next-token probability:

$$\Delta_{\text{sent}} = \frac{1}{|\mathcal{P}|}\sum_{w\in\mathcal{P}} p'(w) - \frac{1}{|\mathcal{N}|}\sum_{w\in\mathcal{N}} p'(w) - \left(\frac{1}{|\mathcal{P}|}\sum_{w\in\mathcal{P}} p(w) - \frac{1}{|\mathcal{N}|}\sum_{w\in\mathcal{N}} p(w)\right), \tag{3}$$

Table 1: Steering efficiency across intervention spaces. Efficiency is defined as sentiment probability shift per unit of KL divergence ($\eta = \Delta_{\text{sent}}/D_{\text{KL}}$). The residual stream is $2.6\times$ more efficient than the MLP intermediate space at their respective optimal α values.

Space	Best Efficiency (η)	Optimal α	Max Δ_{sent}
RESID	**0.0130**	10	**+0.000665**
MLP-OUT	0.0062	10	+0.000248
MLP-INTER	0.0048	5	+0.000243

Table 2: Linear probe accuracy (5-fold CV) for sentiment classification at layer 8. The residual stream achieves the highest accuracy despite having the fewest dimensions. Higher dimensionality does not imply greater linear separability.

Space	Dimensionality	Accuracy
RESID	768	$\mathbf{0.860 \pm 0.102}$
MLP-OUT	768	0.800 ± 0.063
MLP-INTER	3072	0.760 ± 0.163

where $p(w)$ and $p'(w)$ are the baseline and steered next-token probabilities.

KL divergence. We measure $D_{\text{KL}}(p' \| p)$ between the steered and baseline next-token distributions to quantify the magnitude of distributional perturbation, averaged over neutral prompts.

Steering efficiency. We define efficiency as $\eta = \Delta_{\text{sent}}/D_{\text{KL}}(p' \| p)$, which measures the sentiment shift achieved per unit of distributional change. This enables fair comparison across spaces that may require different α values for the same effect.

Linear probe accuracy. We train L2-regularized logistic regression classifiers (5-fold cross-validation) on the collected activations to predict sentiment polarity, measuring how linearly separable the concept is in each space.

Sparsity metrics. We characterize the sparsity of both activations and steering vectors using the Gini coefficient, kurtosis, near-zero fraction (L^0), and top-k concentration (fraction of squared norm in the top $k\%$ of dimensions).

3.6 Multi-Layer Analysis

To verify that our findings generalize beyond layer 8, we repeat the linear probe analysis across layers 6–10, spanning the middle portion of the network.

4 Results

4.1 Steering Efficiency

Table 1 presents the peak steering efficiency for each intervention space. The residual stream achieves the highest efficiency at every operating point, producing $2.6\times$ more sentiment shift per unit of KL divergence than the MLP intermediate space.

All three spaces exhibit an inverted-U relationship between steering strength and effectiveness (Figure 1): sentiment shift increases at moderate α values, peaks, and then reverses at high α (> 100–200) as the intervention pushes activations out of the model's learned distribution. The MLP intermediate space reaches its peak at a lower α (5 vs. 10), consistent with its greater sensitivity to perturbation.

4.2 Linear Separability

Table 2 reports 5-fold cross-validated linear probe accuracy for sentiment classification at layer 8. Despite its $4\times$ higher dimensionality, the MLP intermediate space is *less* linearly separable than the residual stream (76.0% vs. 86.0%), with higher variance across folds.

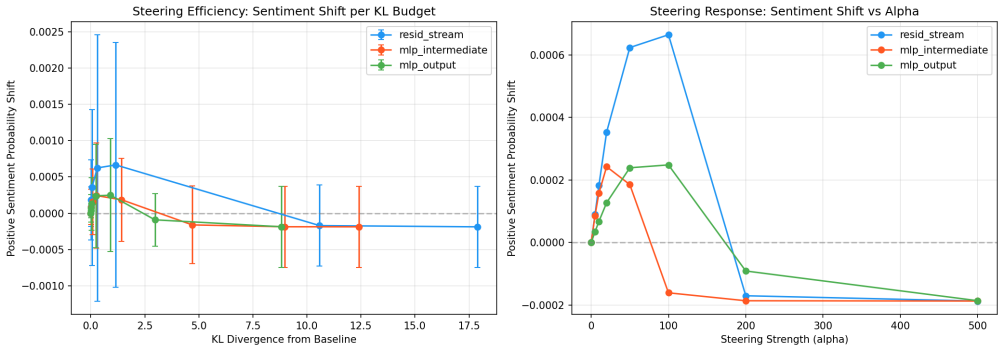

Figure 1: Steering efficiency as a function of intervention strength (α). *(Left)*Sentiment probability shift. *(Right)*Efficiency (shift per unit KL). The residual stream (blue) dominates at all operating points. All spaces show the characteristic inverted-U pattern, with the MLP intermediate space (green) peaking earlier and at a lower magnitude.

Table 3: Linear probe accuracy across layers 6–10. The residual stream advantage is consistent and grows in later layers, where MLP neurons become increasingly specialized.

Layer	RESID	MLP-INTER	MLP-OUT
6	0.820	**0.840**	0.780
7	**0.900**	0.800	0.880
8	**0.860**	0.760	0.800
9	**0.860**	0.720	0.880
10	**0.800**	0.700	0.820

Multi-layer analysis. Table 3 confirms this pattern across layers 6–10. The residual stream is more separable than the MLP intermediate space in 4 of 5 layers, with the gap widening in later layers (layer 10: 80.0% vs. 70.0%). The sole exception is layer 6, where the MLP intermediate space slightly leads (84.0% vs. 82.0%)—possibly because early-layer MLP representations are more distributed and less specialized.

4.3 Sparsity Structure

Steering vector sparsity. The MLP intermediate steering vector is remarkably concentrated (Table 4). Just 86 neurons (2.8% of 3072) account for 50% of the squared norm, and 609 neurons (19.8%) account for 90%. The single most important neuron (index 2002) contributes 4.4% of the steering signal alone. By contrast, the residual stream steering vector requires 43.0% of its dimensions for 90% of the norm—it is far more distributed.

Activation sparsity. Counterintuitively, MLP intermediate *activations* are not sparser than residual stream activations by Gini coefficient (Table 5). The residual stream has a higher Gini coefficient (0.443 vs. 0.223), meaning its activations are more concentrated. However, the MLP intermediate space has substantially higher kurtosis (9.5 vs. 5.4), indicating heavier tails—a few neurons activate very strongly while most remain near their mean. This pattern is consistent with a space containing both many low-level active neurons and a few highly active "expert" neurons.

4.4 Cross-Space Alignment

To understand how the steering vectors relate across spaces, we project the MLP intermediate vector through the down-projection matrix $\boldsymbol{W}_{\text{out}}$ and compute cosine similarities (Table 6).

The perfect cosine similarity between the projected MLP intermediate vector and the MLP output vector is expected—the MLP output is exactly $\boldsymbol{W}_{\text{out}}$ applied to the intermediate activations. The moderate alignment (0.474) between the residual stream vector and the MLP-derived vectors reflects the fact that the residual stream integrates contributions from both attention and MLP, while the MLP vectors capture only the MLP's contribution to sentiment.

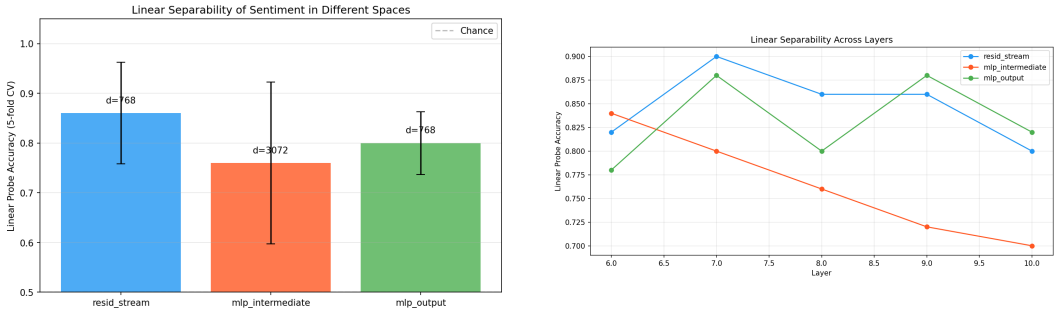

Figure 2: *(Left)*Linear probe accuracy at layer 8 across spaces. *(Right)*Probe accuracy across layers 6–10. The residual stream (blue) is consistently more separable, with the gap widening in later layers.

Table 4: Steering vector concentration. The MLP intermediate steering vector is highly sparse: 2.8% of neurons carry 50% of the signal, compared to 43% of dimensions in the residual stream.

Metric	RESID	MLP-INTER	MLP-OUT
Dims for 50% norm^2	—	**86 (2.8%)**	—
Dims for 90% norm^2	330 (43.0%)	**609 (19.8%)**	336 (43.8%)
Dims for 99% norm^2	—	1648 (53.6%)	—
Normalized entropy	0.954	**0.923**	0.955
SV sparsity (zero frac.)	5.2%	**23.1%**	3.4%

4.5 Qualitative Steering Examples

Table 7 shows steered generations for the neutral prompt "The weather today is" at varying α values. Both spaces produce coherent, positively-steered text at moderate α. At high α, both degenerate—but the MLP intermediate space breaks down at lower α values and in a qualitatively different way (repeating numeric tokens vs. repeating characters).

4.6 Hypothesis Summary

Table 8 summarizes the status of our four hypotheses.

5 Discussion

5.1 Why Is the MLP Space Harder to Steer?

Our results paint a consistent picture: the MLP intermediate space is not "more linear"—it is more *specialized*. Each MLP neuron acts as a feature detector for specific patterns, as evidenced by the high kurtosis of activations (9.5 vs. 5.4 in the residual stream) and the extreme concentration of the steering vector (2.8% of neurons carrying 50% of the signal). When we compute a mean-difference steering vector, the result is a sparse direction that primarily activates a handful of specific neurons. Pushing these neurons produces the desired sentiment shift at moderate strengths, but because the neurons are so specialized, strong interventions quickly drive the model out of distribution.

The residual stream, by contrast, integrates information from all upstream attention heads and MLP sublayers. A single direction in this space corresponds to a broader, more distributed concept representation that degrades more gracefully under perturbation. The 0.474 cosine similarity between residual and MLP-derived steering vectors confirms that the residual stream captures information beyond just the MLP's contribution—attention heads also encode sentiment-relevant features that the MLP vector misses.

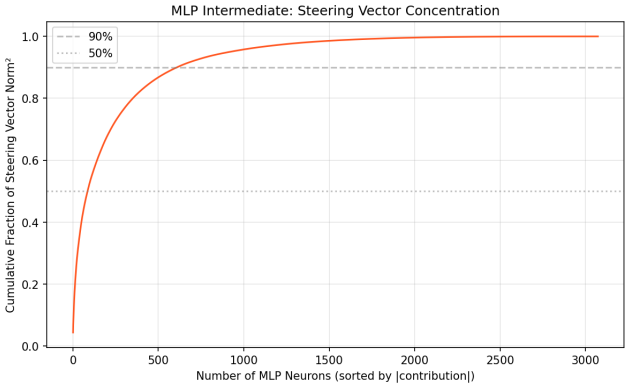

Figure 3: Cumulative squared norm of the MLP intermediate steering vector as a function of the number of neurons (sorted by magnitude). The curve rises steeply: 86 neurons (2.8%) capture 50% of the signal, and the top 609 (19.8%) capture 90%.

Table 5: Activation sparsity metrics. The MLP intermediate space has lower Gini coefficient (less sparse overall) but higher kurtosis (more extreme outliers), consistent with a small number of expert neurons amid many weakly active ones.

Metric	RESID	MLP-INTER	MLP-OUT
L^0 (near-zero frac.)	**0.382**	0.108	0.330
Gini coefficient	**0.443**	0.223	0.329
Kurtosis	5.4	**9.5**	6.6
Top-10% concentration	**0.365**	0.194	0.281

5.2 Sparsity Is a Double-Edged Sword

The sparsity of MLP steering vectors is both a strength and a weakness. On the positive side, the concentration means we can identify a small set of "sentiment neurons" (86 of 3072) that carry most of the steering signal. This could enable more targeted interventions—for example, clamping specific neurons rather than adding a full 3072-dimensional vector. On the negative side, this concentration makes mean-difference steering fragile. A vector that relies on 86 neurons is more sensitive to the magnitude of each component than one that distributes its signal across 330 dimensions.

This connects to the finding of Shafran et al. [2025] that SNMF-derived MLP features outperform direct mean-difference vectors at steering. Their approach identifies *groups* of co-activated neurons that jointly encode a concept, rather than treating each neuron independently. Our results suggest that the mean-difference method is particularly ill-suited to the MLP intermediate space because it does not account for the compositional, hierarchical structure of neuron groups.

5.3 Implications for Steering Methods

Our findings have practical implications for activation steering:

Residual stream remains the best default. For general-purpose steering with mean-difference vectors, the residual stream is strictly better: more efficient, more robust, and easier to work with. The $2.6\times$ efficiency advantage is large enough to matter in practice.

MLP space may excel with feature-based methods. The sparsity and concentration we observe suggest that feature-based steering methods (e.g., SNMF, transcoders) may unlock the potential of the MLP space. Direct mean-difference steering is a blunt instrument for a space that organizes information in sparse, specialized neuron groups.

Targeted neuron interventions deserve exploration. Given that 86 neurons carry 50% of the sentiment steering signal, intervening on just those neurons—with per-neuron scaling calibrated to their baseline activation ranges—could be more effective than full-vector addition.

Table 6: Cosine similarity between steering vectors across spaces. The projected MLP intermediate vector and the MLP output vector are identical (cosine similarity 1.000), validating the methodology. The moderate alignment with the residual stream (0.474) indicates the vectors capture related but distinct aspects of sentiment.

Comparison	Cosine Similarity
RESID vs. projected(MLP-INTER)	0.474
RESID vs. MLP-OUT	0.474
projected(MLP-INTER) vs. MLP-OUT	**1.000**

Table 7: Steered generations for "The weather today is…" at different α values. The MLP intermediate space produces coherent positive text at moderate α but degrades to repetitive tokens at lower intervention strengths than the residual stream.

Space	α	Continuation
RESID	50	"…is good for a good time in the United States."
MLP-INTER	50	"…is a great day to visit. The day is to be a great day to visit."
RESID	200	"…isAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA"
MLP-INTER	200	"…is191919191919191919191919191919"

5.4 Limitations

Single concept. We test only sentiment steering. Other concepts (e.g., factuality, formality, language) may have different sparsity profiles in the MLP space, and our conclusions may not generalize.

Single model. GPT-2 Small uses GELU activations, which produce less sparse intermediate representations than ReLU. Models with ReLU (e.g., early Pythia) or SwiGLU (e.g., Llama) MLPs may show different patterns, and the MLP space could be a better steering target in architectures with sparser activations.

Small dataset. We use 25 contrastive prompt pairs, which may introduce noise in the steering vectors. Larger datasets could yield cleaner vectors, potentially narrowing the efficiency gap.

Greedy decoding. All generations use greedy decoding (temperature 0). Sampling-based decoding might reveal different patterns, particularly at high α values where the distribution is heavily distorted.

Single layer. Our steering experiments focus on layer 8. While the multi-layer probe analysis (layers 6–10) supports generalization, the optimal layer for MLP steering could differ from the optimal layer for residual stream steering.

6 Conclusion

We conducted the first systematic comparison of mean-difference activation steering across three intervention spaces in GPT-2 Small: the residual stream, the MLP intermediate (post-GELU) space, and the MLP output. Our central finding is that the MLP intermediate space can be steered, but is $2.6\times$ less efficient than the residual stream. The wider space is not more linearly separable (76% vs. 86% probe accuracy), and its steering vector is remarkably concentrated: 2.8% of neurons carry 50% of the signal. This concentration makes MLP steering fragile under strong interventions.

These results resolve a gap in the steering literature. While prior work shows that *learned* MLP features (via SNMF or transcoders) can outperform residual-stream methods, we show that the simplest approach—mean-difference steering—is substantially worse in the MLP space. The MLP intermediate space organizes information in specialized neuron groups that resist the broad perturbations

Table 8: Summary of hypothesis testing results.

Hypothesis	Result	Key Evidence
H1: MLP steering produces concept-aligned text	Supported	Positive Δ_{sent} at moderate α
H2: MLP space is more linearly separable	Rejected	Probe: 76% vs. 86% (residual)
H3: MLP space has concentrated concept info	Partial	2.8% neurons $\rightarrow$ 50% signal
H4: MLP steering is more/less reliable	Inconclusive	Similar CV at matched KL

of mean-difference vectors. The residual stream, as an integrative representation, provides a more universal and robust target.

Looking forward, three directions seem promising. First, combining mean-difference vectors with feature-based methods (e.g., steering only the top-k neurons identified by our sparsity analysis) could bridge the gap between simple and learned approaches. Second, testing on models with sparser MLP activations (ReLU or SwiGLU architectures) may reveal settings where MLP steering is more effective. Third, multi-concept analysis would clarify whether the sparsity patterns we observe for sentiment generalize across the space of steer-able concepts.

References

Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. Sparse autoencoders find highly interpretable features in language models. *arXiv preprint arXiv:2309.08600*, 2023.

Jacob Dunefsky, Philippe Chlenski, and Neel Nanda. Transcoders find interpretable LLM feature circuits. In *Advances in Neural Information Processing Systems*, 2024.

Bruce W. Lee et al. Programming refusal with conditional activation steering. *arXiv preprint arXiv:2411.09628*, 2024.

Samuel Marks et al. Language model circuits are sparse in the neuron basis. *arXiv preprint arXiv:2502.08148*, 2026.

Neel Nanda and Joseph Bloom. TransformerLens. *GitHub repository*, 2022.

Neel Nanda, Andrew Lee, and Martin Wattenberg. Emergent linear representations in world models of self-supervised sequence models. *arXiv preprint arXiv:2309.00941*, 2023.

Kiho Park, Yo Joong Choe, and Victor Veitch. The linear representation hypothesis and the geometry of large language models. *arXiv preprint arXiv:2311.03658*, 2024.

Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. Language models are unsupervised multitask learners. *OpenAI Blog*, 2019.

Nina Rimsky, Nick Gabrieli, Julian Schulz, Alexander Matt Turner, et al. Steering Llama 2 via contrastive activation addition. *arXiv preprint arXiv:2312.06681*, 2024.

Or David Shafran, Atticus Geiger, and Mor Geva. Decomposing MLP activations into interpretable features via semi-nonnegative matrix factorization. *arXiv preprint arXiv:2506.10920*, 2025.

Daniel Tan, David Chanin, and Aengus Lynch. Analyzing the generalization and reliability of steering vectors. *arXiv preprint arXiv:2407.12404*, 2024.

Alexander Matt Turner, Lisa Thiergart, David Udell, Gavin Leech, Ulisse Mini, and Monte Cecchetti. Activation addition: Steering language models without optimization. *arXiv preprint arXiv:2308.10248*, 2023.

Daniel Wang et al. Multi-property steering with dynamic activation composition. *arXiv preprint*, 2024.

Zhen Wang et al. Angular steering: Rotation-based steering in activation space. *arXiv preprint*, 2025.

Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, et al. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023.